

Heat shock factors in rice (*Oryza sativa* L.): genome-wide expression analysis during reproductive development and abiotic stress

Harsh Chauhan · Neetika Khurana ·
Pinky Agarwal · Paramjit Khurana

Received: 27 April 2011 / Accepted: 5 July 2011 / Published online: 21 July 2011
© Springer-Verlag 2011

Abstract Plants respond to heat stress by enhancing the expression of genes encoding heat shock protein (HSPs) genes through activation of heat shock factors (HSFs) which interact with heat shock elements present in the promoter of HSP genes. Plant HSFs have been divided into three conserved classes viz A, B and C. In the present study, a detailed analysis has been done of all rice HSFs, along with their spliced variants. Their chromosomal localization reveals that six HSFs are segmentally duplicated and four pairs of these segmentally duplicated HSF encoding genes show pseudo-functionalization. Expression profiling through microarray and quantitative real-time PCR showed that eight *OsHsf*s express at a higher level during seed development, while six HSFs are up-regulated in all the abiotic stresses studied. The expression of *OsHsfA2a* gene in particular was greatly stimulated by heat stress in both root and shoot tissues and also during panicle and seed development. *OsHsfA3* was found more responsive to cold and drought stress, while *OsHsfA7* and *OsHsfA9* showed developing seed-specific expression. This study also revealed that spliced variants generally accumulated at a higher level in all the tissues examined. Different hormones/elicitors like ABA, brassinosteroids and salicylic acid also alter *OsHsf* gene expression. Calcium in

combination with heat stress elevated further the level of HSF transcripts. Expression analysis by both microarray and real-time RT-PCR revealed a unique stable constitutive expression of *OsHsfA1* across all the tissues and stresses. A detailed in silico analysis involving identification of unidentified domains has been done by MEME-motif tool in their full-length proteins as well as in DNA-binding domains. Analysis of 1 kb putative promoter region revealed presence of tissue-specific, abiotic stress and hormone-related *cis*-acting elements, correlating with expression under stress conditions.

Keywords Abiotic stress · Heat shock factor · Rice · Transcription profiling

Introduction

Plants cannot escape from unfavorable environmental conditions and thus have developed special measures to cope with stressful conditions. This is generally achieved by altered gene expression mediated by stress-responsive transcription factors such as heat shock factors. Plants show different but overlapping expression pattern for different abiotic stresses which sometimes occur simultaneously in the field conditions such as drought and heat stress (Mittler et al. 2001; Rizhsky et al. 2002). Heat shock response (HSR) is a universal conserved phenomenon, displayed by all eukaryotes. In plants, HSR is characterized by perception of the signal at the membrane (Saidi et al. 2009) and expression of heat shock protein (HSP) genes. These HSPs function as molecular chaperones (Schoffl et al. 1998) and their expression is governed by heat shock factors (HSFs) through their interaction to heat shock elements (HSE; nGAAn pentamers) present in the promoter

Communicated by P. Westhoff.

Electronic supplementary material The online version of this article (doi:10.1007/s00438-011-0638-8) contains supplementary material, which is available to authorized users.

H. Chauhan · N. Khurana · P. Agarwal · P. Khurana (✉)
Department of Plant Molecular Biology, University of Delhi,
South Campus, Benito Juarez Road, Dhaulau Kuan,
New Delhi 110 021, India
e-mail: param@genomeindia.org

region (Nover et al. 2001). Unlike other eukaryotes, which have one to four members of the HSF family, plants characteristically have large number of HSFs in their genomes; *Arabidopsis* and rice have 21 and 25 members, respectively (Nover et al. 2001; Guo et al. 2008a). Based on sequence homology and domain architecture, plant HSFs have been divided into three conserved classes. While A type HSF are characterized by exclusive presence of AHA transactivator domain in their C-terminal transactivation domain (CTD), class B and class C HSFs do not possess a trans-activator domain. The class B HSFs are more similar to non-plant HSFs than class A and C, as these have a longer HR-A/B region due to insertion of 21 and 7 residues, respectively (Nover et al. 2001; Kotak et al. 2004). Plant HSFs are represented by conserved structural features such as presence of highly conserved N-terminal, helix turn helix type DNA binding domain, followed by an oligomerization domain composed of two hydrophobic heptad repeats (HR-A and HR-B) connected to DBD by a linker of 15–80 residues and presence of nuclear localization (NLS) and nuclear export signals (Nover et al. 2001). In contrast to N-terminal architecture, the CTD, possessing AHA motif, is less conserved in plants.

The involvement of HSFs in various abiotic stresses is well studied (Charnng et al. 2007; Sakuma et al. 2006; Schramm et al. 2008; Yoshida et al. 2008) and ectopic expression of rice HSF was found to increase tolerance to abiotic stress in transgenic *Arabidopsis* (Yokotani et al. 2008). There is great diversity in the function of plant HSFs in the HSR. Interactions of HSF with HSPs and other transcription factors have been investigated (Nover et al. 2001) and it has been found that constitutive expression of tomato HsfA1a is central for HSR (Mishra et al. 2002). The role of HsfA2 and A3 in HSR was also elucidated in tomato response to heat stress (Scharf et al. 1998; Bharti et al. 2000). Apart from heat stress, plant HSFs are also induced by other abiotic stresses such as cold, salt and drought and overlapping response of different HSFs and stress environment were observed in *Arabidopsis* (Swindell et al. 2007). In rice *spl17* mutant is defective in *OsHsfA4d*, which leads to the disease mimic spotted leaf phenotype under high temperature (Yamanouchi et al. 2002). Another rice HSF gene *OsHsfA4a* is involved in cadmium tolerance (Shim et al. 2009). In *Arabidopsis*, over-expression of HsfA4a leads to decreased expression of cytosolic H₂O₂ scavenging ascorbate peroxidase and it was hypothesized that HSFs may act as H₂O₂ sensors in the plants (Davletova et al. 2005; Miller and Mittler 2006).

In the present investigation, we carried out genome-wide transcription analysis of all rice HSFs together with their spliced variants, in an array of 28 different rice tissues representing major growth stages and abiotic stress conditions involving microarray and quantitative RT-PCR

analysis. Additionally, inducibility of different rice HSFs by secondary messengers and early heat stress response was elucidated. Analysis of duplicated HSF genes and their expression was also undertaken.

Materials and methods

Rice HSFs

The nucleotide and protein sequence of rice HSFs and their predicted alternative spliced variants were taken from Plant Transcription Factor Database (Pln TFDB) and reconfirmed by BLASTN and BLASTP in TIGR-Rice database. They were then assigned the chromosomal location on rice chromosome pseudomolecules by the BLASTN search. Phylogenetic tree was constructed using CLUSTALX (1.83) program and an un-rooted neighbor-hood joining method based phylogenetic tree with bootstrap values was constructed with default parameters under multiple alignment. The phylogenetic trees were displayed using TreeView (version 1.6.6, <http://www.taxonomy.zoology.gla.ac.uk/rod/treeview.html>). The 1-kb up-stream region of translation start codon was analyzed for the presence of putative cis-acting elements by ELEMENT motif search tool (Mockler et al. 2007) which calculates the Z score and the *p* value along with inbuilt connections to PLACE (Higo et al. 1998) and PLANTCARE (Rombauts et al. 1999). Presence of putative NLS was predicted by PredictNLS (<http://cubic.bioc.columbia.edu/cgi/var/nair/resonline.pl>) and putative nuclear export signal was searched by NetNES 1.1 (<http://www.cbs.dtu.dk/services/NetNES/>). Domain search program SMART (<http://smart.embl/heidelberg.de/>) was used to deduce the domain structure of putative HSF protein sequences. Additional conserved motifs in HSF full protein sequences as well as in DNA-binding domain were identified by MEME motif search tool. All rice HSFs or their respective DNA-binding domains were entered as input and a limit of 20 motifs was specified with the default parameters. ProScan (<http://genamics.com/expression/proscan.htm>) was used to find the biologically relevant protein signatures in the motifs identified by MEME.

Localization on rice chromosomes and segmental duplication

Each of the *OsHsf* genes was positioned on rice chromosome pseudomolecules available at TIGR (release 5) (<http://www.tigr.org/tldb/e2k1/osa1/pseudomolecules/info.shtml>) and IRGSP (build 05) (<http://rgp.dna.affrc.go.jp/E/IRGSP/Build5/build5.html>) by BLASTN search. Segmental duplication of rice genes available at TIGR

(http://www.tigr.org/tdb/e2k1/osa1/segmental_dup/index.shtml) was used to determine the presence of HSF genes on duplicated chromosomal segments, with the maximal length distance permitted between collinear gene pairs of 500 kb.

Full-length cDNA and EST evidence search

The FL-cDNAs and paired ESTs were searched at KOME-FL cDNA database as well as the gene expression evidence search page (http://www.tigr.org/tdb/e2k1/osa1/locus_expression_evidence.shtml) available at TIGR rice genome annotation. TIGR locus IDs corresponding to the rice HSF genes were searched to find the availability of corresponding full-length cDNA sequences. For EST-based expression analysis, BLAST search was performed using the cDNA sequences of rice HSF genes against the EST database (<http://www.ncbi.nlm.nih.gov/dbEST/index.html>) available at NCBI. To obtain information on the intron/exon structures, cDNA sequences of Hsf genes were aligned with their corresponding genomic sequences.

Plant material and growth conditions

Rice, *Oryza sativa* (sp. *indica* var. IR64), seeds were obtained from Indian Agriculture Research Institute, New Delhi, for the present study. Plant material for different developmental stages and their growth conditions for microarray analysis are the same as described previously (Nijhawan et al. 2008). For stress treatments, 7-day-old-light grown seedlings were dried for 3 h (dehydration stress), treated with 200 mM NaCl for 3 h (salt stress), kept at $\pm 4^{\circ}\text{C}$ for 3 h (cold stress) and kept at 37°C for 3 h (heat stress). Similarly, ABA (10 μM , Sigma, USA), brassinosteroid (1 μM epibrassinolide, Sigma) and salicylic acid (100 μM) were also given for 3 h. For short-term treatments, seedlings were either kept for 15 min at 37°C for heat stress, or CaCl_2 (10 mM) for 15 min, and for combined calcium and heat stress, seedlings were immersed in 10 mM CaCl_2 solution at 37°C . After treatment, sampling was done and tissue frozen and stored at -80°C .

RNA isolation

Total RNA from shoot and root tissues was extracted using the RNeasy Plant Mini Kit (Qiagen, Germany) including on-column treatment with RNase-free DNaseI (Qiagen) to remove any genomic DNA contamination according to the manufacturer's instructions. The integrity of RNA samples was monitored by electrophoresis on 1.2% agarose gel. For each RNA sample, the concentration was measured by Nanoview (GE Healthcare, UK). To isolate RNA from mature and immature seeds, the RNA isolation method

used for wheat and other carbohydrate-rich seeds was used (Singh et al. 2003), followed by treatment with DNase I to remove genomic DNA contamination.

Microarray analysis

Microarray analysis was performed using ArrayAssist 5.0.0[®] on our in-house available data of vegetative (7-day-old seedling), reproductive (P4, P5, P6, S1, S2, S3, S4 and S5) and abiotic stress (cold, drought and salt) stages [Gene Expression Omnibus (GEO) platform accession number GPL2025]. The experiment has been described in detail previously (Agarwal et al. 2007; Nijhawan et al. 2008). Also, analysis was done on heat stress from publicly available data (GEO dataset accession number GSE14275). Microarray analysis involved hierarchical clustering of normalized and $\log_2$ transformed data to generate a heat map. The algorithm used Squared Euclidean distance metric and Ward's linkage rule for clustering. Differential expression analysis was performed on cold, drought and salt stress and reproductive stages in comparison with 7-day-old seedling and 14-day-old seedling heat stress stages. Benjamini and Hochberg (1995) correction was applied to the reproductive stages only. The signal values after normalization were also used to compare the expression patterns of duplicated genes.

Quantitative real-time PCR analysis

Transcript levels for all rice HSF genes as well as their respective spliced variants in different RNA samples were quantified by real-time PCR analysis using the ABI Prism 7000 Sequence Detection System and Software (PE applied Biosystems, USA) as per manufacturer's instructions. The qRT-PCR was done in an array of 28 different tissues (Table 2) representing different plant developmental stages as well as abiotic stresses. Primers were designed from unique regions of the genes using Primer Express 2.0 (Applied Biosystems, USA; Supplemental Table S1 in Supplementary Material). For spliced variants, at least one primer was designed from the cDNA which takes part in alternative splicing. To ensure that the primers amplify a unique and specific fragment of the cDNA, each primer pair was checked by homology searches against the rice genomic sequence at the TIGR Web site. First-strand cDNA was synthesized by reverse transcription of 2 μg of total RNA using High Capacity cDNA Archive kit (Applied Biosystems, USA) as per manufacturer's instructions. These cDNA samples were mixed with 200 nM of each primer and SYBR Green PCR Master Mix (Applied Biosystems, USA) in 96-well optical reaction plates (Applied Biosystems, USA). The identity of the

amplicons and the specificity of the reaction were verified by melting curve analysis. The relative mRNA levels for each of the HSF genes in different RNA samples were normalized with respect to the internal standard *actin*, to normalize for variance in the quality of RNA and the amount of input cDNA. Real-time PCR analysis was based on two biological replicates of each sample and three technical replicates of each biological replicate (Chaudhary et al. 2009).

Results

Rice HSF gene family

The plant transcription factor database (PlnTFDB 3.0) (Perez-Rodriguez et al. 2009) revealed 25 genes as *OsHsf*, similar to that reported by Guo et al. (2008b). Of these, 13 genes belong to Class A, 8 genes to Class B and remaining 4 to Class C type HSF (Table 1). The characteristic features, protein length, DNA-binding domain length, presence of introns, NLS and nuclear export sequence are given in Table 1. Rice FL-cDNA for at least 22 HSF genes could be found, while *OsHsfA2f*, *OsHsfB4a* and *OsHsfB4d* do not have full-length cDNAs in KOME database and these are not represented by paired ESTs also. All the rice HSF genes have introns, most of them having only one. All the class A HSFs are highly expressed in terms of presence of ESTs in the database, as all are represented by more than 20 ESTs except *OsHsfA2f* and A7, having 9 ESTs each. Among all the HSFs, 12 have clear predicted NLS. All class A HSFs have NLS except for *OsHsfA2f*, A3, A4 and A5 and only one class B HSF, *OsHsfB1*, has a predicted NLS. Two of the four class C HSFs also possess NLS (Table 1). Six rice HSFs have clear predicted NES, involving three class A and three class B HSFs. Interestingly, one spliced variant of *OsHsfA2e* (LOC Os03g58160.2) has an NES, while the main LOC Os03g58160.1 does not have the predicted NES (Table 1).

The rice HSF family is distributed on 10 of the 12 chromosomes; no HSF gene is present on chromosome 11 and 12. Maximum six HSF genes are present on chromosome 3 (Fig. 1). The position and direction of transcription (arrows) of each HSF gene are marked with the help of International Rice Genome Sequencing Project (IRGSP) rice chromosome pseudomolecules (Fig. 1). Six of the 25 rice HSFs are found to be segmentally duplicated and in three cases the orientation is also reversed. Both the HSFs found on chromosome 8 are segmentally duplicated with 2 of the HSF on chromosome 9, and all are class B HSF. In total, three class B, two class A and one class C HSF are segmentally duplicated. The single HSF present on chromosome 10 is also duplicated (Fig. 1).

Phylogenetic analysis

To explore the evolution of the OsHSFs in the rice genome, an un-rooted tree was constructed from alignments of their full-length protein sequences (Fig. 2). It is seen that rice HSFs are clearly grouped into three major clades as per their A, B and C type with well-supported boot strap values. All class A HSFs are grouped in one single major clade with two distinct sub groups of A2, A3, A7 and A5, A4, A9 and A1. All class B HSFs also grouped together and appear to have diverged from the type A subgroup having A1 in close proximity. Class C HSFs also make different group and appear closer to *OsHsfA7*. Therefore, it appears from the phylogenetic tree that one subgroup of class A HSF gave rise to class B and the other to class C. As expected, all the duplicated HSFs grouped together in the phylogenetic tree (Fig. 2).

Analysis of conserved unidentified domains

All the rice HSFs were analyzed by MEME motif search tool to identify conserved motifs in their full-length proteins as well as motifs in the respective DNA binding domains. There is great diversity in the C terminal ends of full-length proteins of various HSFs (Supplementary Fig. 1). Motif 11, 12, 13, 14 and 18 are exclusively present in class A HSFs only, which together may represent the activation domain. Additionally, motif 16 is also present in class A HSFs only, except for one B-type HSF B4A. Similarly, motif 10 is also present in class A HSFs only, except for one class C HSF, C1a. Motif 15 is also found in *OsHsfA2a* and *OsHsfB2c*. Both these HSFs show highest expression during heat stress by quantitative RT-PCR analysis (Table 2). ProScan analysis revealed that many of these motifs possess phosphorylation sites such as protein kinase C and casein kinase II (Supplementary Table 2).

As far as the DBD domain analysis is concerned, we again find greater diversity towards the C-terminal end (Supplementary Fig. 2). Motif 10 is present in only class A HSFs; however, not all HSFs of this class harbor motif 10. One class C HSF, C1b, also has motif 10. This motif is altogether absent from all class B HSFs. Similarly, motif 4 is also absent from class B HSFs. Interestingly, when present in class A HSFs, motif 10 is flanked by motif 6 on either side except in case of *OsHsfA4b*.

Microarray-based expression analysis

Unique probe sets for all 25 rice HSFs were found on the Rice Affymetrix GeneChip® and their expression was studied in 7-day-old seedling, panicle development stages P4, P5, P6 (corresponding to 10, 20 and 30 cm spike length) and five seed development stages S1–S5

Table 1 Organization of rice HSF genes

Locus ID	<i>OsHsf</i>	FL cDNA	ESTs in TIGR database	Protein length	DBD/length	Exon/introns	NLS/site	NES/site
Os03g63750.1	<i>OsHsfA1a</i>	AK100430	>20	500	36–210 (175)	1	NKKRRPKQ (235) KKRRPK (236)	–
Os03g53340.1	<i>OsHsfA2a</i>	AK069579	>20	372	52–231 (180)	1	LKTIKRR (135)	–
Os03g53340.2			>20	372	52–231 (180)	1	LKTIKRR (135)	–
Os03g53340.3			>20	372	52–231 (180)	2	LKTIKRR (135)	–
Os03g53340.4			>20	285	21–143 (123)	1	LKTIKRR (48)	–
Os07g08140.1	<i>OsHsfA2b</i>	AK101284	15	368	51–219 (169)	1	LKNIKRR (134) KKRRRR (244) KRRRRI (245)	–
Os10g28340.1	<i>OsHsfA2c</i>	AY344488	>20	354	41–210 (170)	1	LKTIKRR (124)	–
Os10g28340.2			>20	354	41–210 (170)	3	LKTIKRR (124)	–
Os10g28340.3			>20	354	41–210 (170)	3	LKTIKRR (124)	–
Os10g28340.4			>20	255	2–110 (109)	1	LKTIKRR (25) KRKELEDAISKRRRR(127)	–
Os03g06630.1	<i>OsHsfA2d</i>	AK066844	>20	375	36–224 (189)	2	LKMIKRR (139)	–
Os03g06630.2			>20	355	36–204 (169)	1	LKMIKRR (119)	–
Os03g58160.1	<i>OsHsfA2e</i>	AK068660	>20	353	37–204 (168)	1	LKSIKRR (120) KKRRR (230) KRRRRI (231)	–
Os03g58160.2			>20	194	37–197 (161)	2	LKSIKRR (120)	LKRDHLLM (157)
Os06g36930.1	<i>OsHsfA2f</i>	nil	nil	327	42–208 (167)	1	–	LTFEALAL (243)
Os02g32590.1	<i>OsHsfA3</i>	AK101934	>20	492	66–230 (165)	1	–	–
Os02g32590.2			>20	445	66–230 (165)	2	–	–
Os01g54550.1	<i>OsHsfA4b</i>	AK109856	>20	435	11–176 (166)	1	–	–
Os05g45410.1	<i>OsHsfA4d</i>	AK100412	>20	454	19–187 (169)	1	–	–
Os02g29340.1	<i>OsHsfA5</i>	AK065643	>20	470	19–186 (168)	1	–	–
Os01g39020.1	<i>OsHsfA7</i>	AK064271	9	398	91–262 (172)	1	LKNIKRR(176) KRRRV (180)	–
Os03g12370.1	<i>OsHsfA9</i>	AK072571	>20	405	49–225 (177)	3	LKTIKRR (131)	LKRDKALLMQ (177)
Os03g12370.2			>20	401	49–225 (177)	1	LKTIKRR (131)	LKRDKALLMQ (177)
Os03g12370.3			>20	401	49–225 (177)	2	LKTIKRR (131)	LKRDKALLMQ (177)
Os09g28354.1	<i>OsHsfB1</i>	AK101182	>20	299	28–224 (197)	1	GRRGGGGGG (18)	VRQLDLRL (209)
Os04g48030.1	<i>OsHsfB2a</i>	AY344483	11	302	9–213 (205)	1	–	LSDLVLAL (272)
Os08g43334.1	<i>OsHsfB2b</i>	AK101700	>20	609	190–400 (211)	6	–	–
Os08g43334.2			>20	386	47–257 (211)	1	–	–
Os09g35790.1	<i>OsHsfB2c</i>	AK106545	>20	449	37–268 (232)	2	–	–
Os09g35790.2			>20	409	37–268 (232)	1	–	–
Os09g35790.3			>20	102	37–102 (65)	1	–	–
Os08g36700.1	<i>OsHsfB4a</i>	nil	nil	376	23–261 (239)	1	–	–
Os07g44690.1	<i>OsHsfB4b</i>	AY344494	>20	307	32–247 (216)	1	–	–
Os09g28200.1	<i>OsHsfB4c</i>	AK241190	14	390	22–264 (243)	1	–	LECDDL SL (361)
Os03g25120.1	<i>OsHsfB4d</i>	nil	nil	302	34–249 (216)	1	–	–

Table 1 continued

Locus ID	<i>OsHsf</i>	FL cDNA	ESTs in TIGR database	Protein length	DBD/length	Exon/introns	NLS/site	NES/site
Os01g43590.1	<i>OsHsfC1a</i>	AK069479	2	344	25–217 (193)	1	RKRRR (235)	–
Os01g43590.2			4	335	25–208 (184)	1	RKRRR (226)	–
Os01g53220.1	<i>OsHsfC1b</i>	AK066316	17	247	23–176 (154)	1	KRRRI (206)	–
Os02g13800.1	<i>OsHsfC2a</i>	AK106488	>20	295	13–177 (165)	1	–	–
Os06g35960.1	<i>OsHsfC2b</i>	AK241254	27	275	10–175 (166)	1	–	–

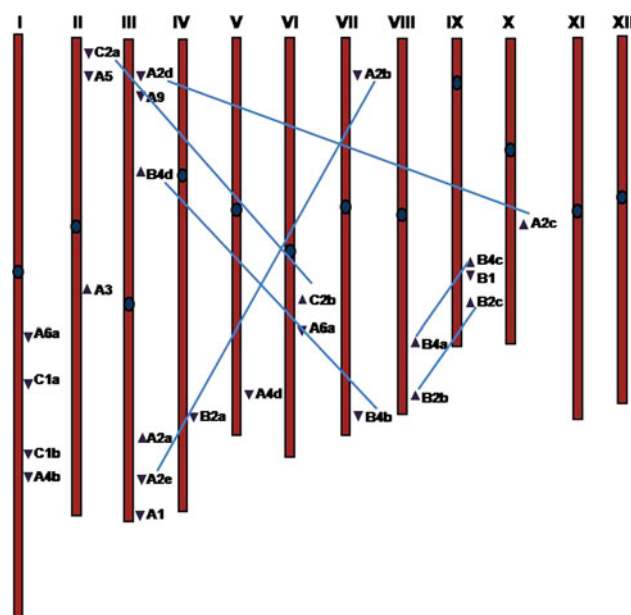


Fig. 1 Chromosomal localization of rice HSF genes. 25 HSF genes were mapped on different rice chromosomes according to 5' and 3' coordinates mentioned in TIGR. Respective chromosome numbers are written at the top. Genes lying on duplicated segments of rice genome have been joined by lines. Arrow indicates the direction of the gene

(corresponding to 0–2, 3–4, 5–10, 11–20 and 21–29 DAP, respectively) (Agarwal et al. 2007). Additionally, the expression data of genes expressing differentially during abiotic stresses, such as cold, drought, salt and heat stress were also considered for analysis of rice HSFs. On the basis of expression in different tissues, a heat map was generated and all rice HSF biosynthesis genes were grouped into two main clades: the first group of genes had moderate to high level of expression, and the second group had low to moderate level of expression in all vegetative, reproductive stages and abiotic stresses studied (Fig. 3). Out of 25 *OsHsfs*, 22 genes showed more than 1.5-fold change in at least one of the tissue taken in this study (Fig. 3 and Supplementary Table 3). Expression of 20 HSFs was found in at least one of the flower or seed development stages. HSFs showing more than twofold change in at least one of the flower development stages are A2a, A2c, A2d, A7, C2a and C2b. *OsHsfB1* shows high expression during later stages of flower development, i.e. 30 cm (P6) and maintains high expression in all the subsequent seed development stages. 10 *OsHsfs* show expression in almost all stages of seed development viz. A2a, A2c, A3, A7, A9, B1, B2c, C1b, C2a and C2b. Importantly, *OsHsfA2a* and A7 have exceptionally high values of fold change in almost all stages of seed development (Supplementary Table 3). Interestingly, both these HSFs have high expression during P4 stage of flower development. Five *OsHsfs*, namely, A2a, A2d, A7, C2a and

Table 2 Expression analysis (relative fold change) of rice HSF gene family in different plant growth stages and various abiotic stresses

HSF	Root			Shoot				Pannicle (CM)				Developing seed (DAP)												
	HS	NaCl	Drought	ABA	BR	SA	Cold	HS	NaCl	Drought	ABA	BR	SA	Cold	(P4) (P5) (P6)				0-2 (S1)	2-4 (S2)	5-10 (S3)	11-20 (S4)	21-30 (S5)	
Oso03g63750.1/A1a	2	-	-	3	-2	-2	2	-	2	-	2	-	2	-	-	10	20	30	2	2	2	2	4	5
Oso03g53340.1/A2a	130	5	-	5	2	-	-	284	3	9	6	2	8	2	64	-	534	5	-14	118	4	4	2	
Oso03g53340.3/A2a	239	15	-	16	2	-	-	1,342	8	10	3	-	15	-	98	6	982	22	3	437	72	199		
Oso03g53340.4/A2a	6,837	206	18	87	26	2	8	4,263	11	12	9	2	30	-	296	15	3,303	30	2	544	167	163		
Oso07g08140.1/A2b	-	-	8	8	-	-	-	-	-	4	-	-	-	-7	-	3	2	-14	-15	-4	-4	-5		
Oso10g28340.1/A2c	42	4	-	12	2	-	-2	56	9	8	7	3	5	-2	23	10	139	2	2	54	30	21		
Oso10g28340.4/A2c	51	5	2	13	3	2	-2	50	11	8	6	2	5	-2	29	10	141	2	-	48	13	10		
Oso03g06630.1/A2d	134	6	3	2	3	2	8	158	12	2	3	2	4	2	7	15	69	9	3	57	6	5		
Oso03g06630.2/A2d	6,258	-	113	10	5	5	10	2,627	28	11	13	7	8	6	17	11	113	15	6	266	57	16		
Oso03g58160.1/A2e	3	-	-	3	-	-4	3	5	2	2	3	2	-	-2	2	3	8	-3	-28	3	-	-2		
Oso03g58160.2/A2e	7	2	-	12	2	2	6	12	2	-	2	-	2	-2	-	-	3	-2	-2	-	-2	-		
Oso06g36930.1/A2f	7	-	-3	2	-2	-	-	47	6	3	3	-	5	-	-	13	32	9	4	14	10	14		
Oso02g32590.1/A3	4	5	3	2	-	2	80	17	7	6	3	-	6	50	6	12	47	4	4	10	5	6		
Oso02g32590.2/A3	5	15	3	2	-	2	79	11	7	6	3	-	4	46	3	17	39	5	5	6	3	6		
Oso01g54550.1/A4b	-3	-	-5	2	-	-2	4	-	3	2	3	2	3	3	3	4	8	3	2	5	2	4		
Oso05g45410.1/A4d	-3	2	-	-	-2	-3	-	-	6	3	2	2	7	-	-	-	-	7	2	2	2	-		
Oso02g29340.1/A5	-	-	-9	2	-2	-3	2	2	4	4	3	2	4	2	3	3	10	3	3	7	5	4		
Oso01g39020.1/A7	5	2	6	5	7	5	3	2	3	3	3	2	2	3	4	3	-	2	-	5	24	20		
Oso03g12370.1/A9	-2	-2	-4	-	-2	-5	2	2	-	2	2	2	3	8	2	4	6	8	-	7	7	-3		
Oso03g12370.2/A9	-2	-	-2	2	-	-3	5	5	8	3	6	3	7	6	3	6	11	10	4	14	9	8		
Oso03g12370.3/A9	-	-	-	2	-	-5	5	5	8	3	5	2	8	5	2	4	3	5	5	15	-	4		
Oso09g28354.1/B1	2	-2	-	2	-	-	-	9	7	4	5	3	6	-	-	18	41	7	3	4	7	4		
Oso04g48030.1/B2a	256	25	3	13	4	3	3	124	13	10	9	5	9	3	12	20	160	3	3	42	24	8		
Oso08g43334.1/B2b	5	6	2	11	-	-10	3	15	7	-	3	-	8	-2	9	2	13	47	23	22	38	15		
Oso08g43334.2/B2b	171	20	3	11	4	3	3	143	13	11	7	4	8	-	15	8	182	2	3	29	22	28		
Oso09g35790.1/B2c	22	43	12	35	9	3	7	194	69	76	24	12	69	2	160	16	27	40	10	58	54	67		
Oso09g35790.2/B2c	81	11	3	13	5	3	2	254	42	35	12	9	26	-	56	30	83	9	3	18	16	24		
Oso08g36700.1/B4a	2	-	-	-	2	2	-	-	-	2	-	-	2	-	-	-	-	-	2	2	-	-		
Oso07g44690.1/B4b	-	2	-	3	-	2	3	-	5	5	5	2	3	-	6	2	2	3	-	-	3	6		
Oso09g28200.1/B4c	-2	-	-	3	-	-	-	-	2	2	4	3	2	-2	8	-	2	-	-9	-	-	3		
Oso03g25120.1/B4d	-	-	-	6	-	4	4	-4	4	2	4	-	2	-	6	2	2	3	-3	-	3	6		
Oso01g43590.1/C1a	9	2	2	8	2	-	5	17	12	18	11	6	6	8	9	8	17	9	5	35	16	17		
Oso01g43590.2/C1a	-2	-3	-6	2	-6	-8	-3	3	5	4	2	-	2	2	5	2	24	2	-	10	4	6		
Oso01g53220.1/C1b	-11	-5	-7	-	-10	-12	-4	-	-	-	2	-	-	3	-6	-	-	-2	-3	-2	-	-		

Table 2 continued

HSF	Root				Shoot				Pannicle (CM)				Developing seed (DAP)									
	HS	NaCl	Drought	ABA	BR	SA	Cold	HS	NaCl	Drought	ABA	BR	SA	Cold	10 (P4)	20 (P5)	30 (P6)	0-2 (S1)	2-4 (S2)	5-10 (S3)	11-20 (S4)	21-30 (S5)
Os02g13800.1/C2a	11	6	7	-4	15	36	4	2	8	2	-2	8	3	2	8	-2	-2	4	-	2	2	-
Os06g35960.1/C2b	-3	2	2	8	-7	-3	2	-3	2	2	2	-	-	4	6	2	4	4	2	2	4	2

Values are relative fold change in qRT-PCR and—represents less than 1.5 fold change

Values are relative fold change in qRT-PCR and—represents less than 1.5 fold change

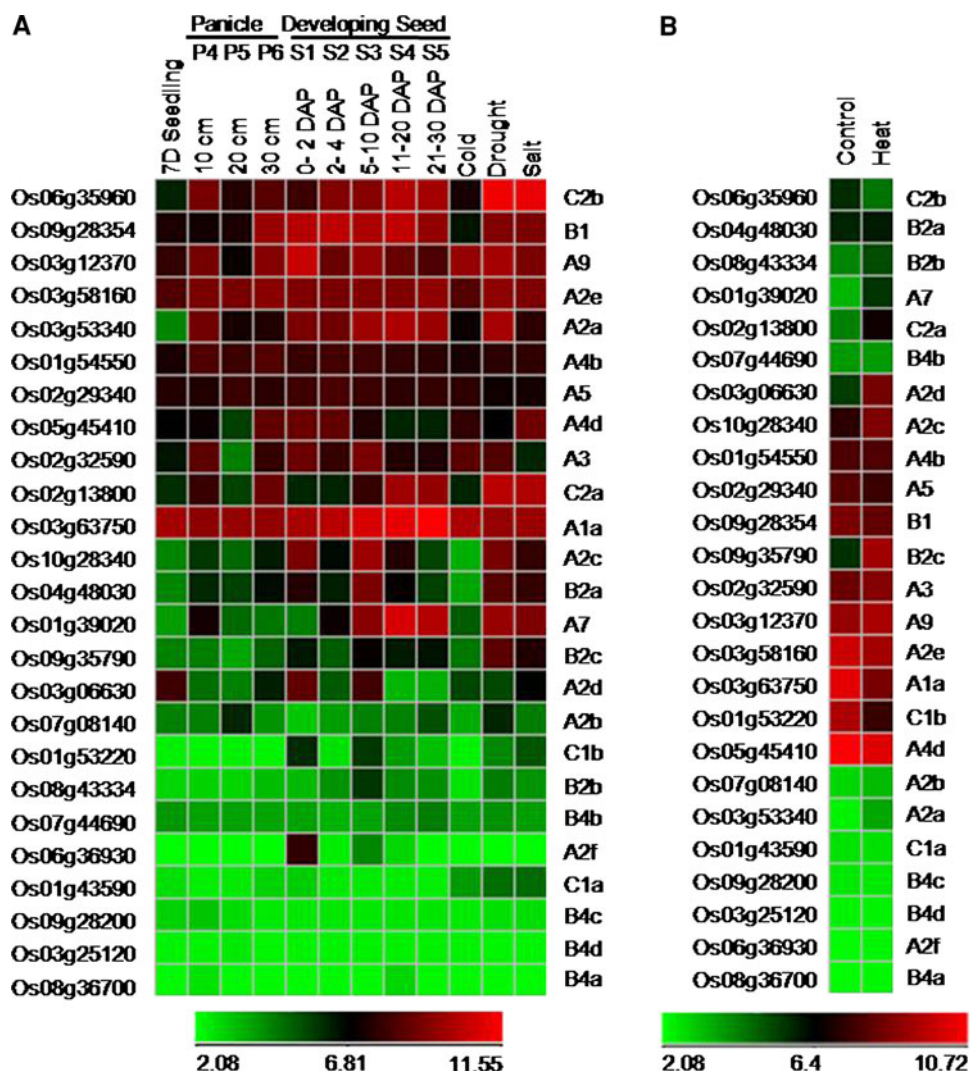
A3 N A9, B2a, B2c and C2b. Particularly A2a and B2a showed very high expression during mid stages of seed development. Five HSFs (*OsHsfA1a*, *A4b*, *A4d*, *A5* and *B1*) show moderate expression in at least some stages of seed development.

As far as abiotic stresses are concerned, six HSFs are up-regulated under all the abiotic stresses studied in the present investigation. *OsHsfA2a*, *A2d*, *B2a*, *B2b*, *B2c* and *C2a* are present during all the abiotic stresses both in root and shoot tissues. *OsHsfA2c* was also induced by all the stresses in the shoot tissues but only by heat stress in root tissue. *OsHsfA3* is particularly induced by cold stress in both root and shoot tissues. Salt stress induced three genes viz. *OsHsfA2a*, *B2a* and *B2b*. A similar number is also induced by drought stress namely, *OsHsfA2d*, *B2c* and *C1a*; however, *C1a* is induced in shoot tissue only.

Hormones play an important role in stress signal transduction and subsequent cell response in terms of altered transcription. ABA is found to regulate nearly all the *OsHsf* genes, except three HSFs, namely *A4d*, *B4a* and *C1b*. Surprisingly, *OsHsfC2a* is the only HSF which is down-regulated by ABA in both root and shoot tissues. Brassinosteroids (BR) also regulate gene expression and we found seven *OsHsf*s (*A2a*, *A2c*, *A2d*, *B2a*, *B2b*, *B2c* and *C1a*) to be up-regulated by BR; all these genes are induced by heat stress also. Nine of the *OsHsf* genes (*A1a*, *A2f*, *A4*, *A5*, *A7*, *A9*, *C1b* and *C2b*) are down-regulated by BR treatment particularly in the root tissues. Salicylic acid (SA) also affects gene expression differentially. SA down-regulates all the *OsHsf* genes in root tissues, except *A2a*, *A2d*, *A3*, *B2b*, *B2c*; all these HSFs are largely involved in temperature stress, either high or low temperature. *OsHsfC2a* is the only HSF, which is highly up-regulated in the root tissues; however, it is the only HSF down-regulated by ABA in both root and shoot tissues. In shoot tissues, all the *OsHsf* genes are up-regulated by SA, particularly *A2a* and *B2c*. Both these genes also have exceptionally high expression levels during heat stress in both root and shoot tissues.

We have also analyzed the number of introns in duplicated gene pairs and found that except for two cases the number of introns is same in duplicated genes. We monitored the expression of duplicated gene pairs by real-time RT-PCR in relation to number of introns in the genomic clones. The gene *OsHsfA2d*, which is duplicated with *OsHsfA2c*, has two introns in place of one in the original gene *A2c*. Interestingly, it showed around 50 times more expression in terms of relative fold change, particularly during heat stress and considerably higher expression in almost all the other stresses and during developing seed tissues (Table 2). The other pair, in which we found different number of introns is duplicated gene pair *OsHsfB2b* and *B2c* (Table 1). Here also, *OsHsfB2c* having 2 introns,

Fig. 3 Hierarchical clustering of rice HSF genes during reproductive development and abiotic stress. Squared Euclidean distance metric and Ward's linkage rule were used for the analysis. The gene IDs are on the left of the heat map while the gene names are on the right. The bars showing the range of $\log_2$ signal values for both heat maps are also shown. **a** The tissues considered for analysis were 7-day-old seedling (7SDL), panicle development stages (at 10, 20 and 30 cm lengths), seed development stages (0–2, 3–4, 5–10, 11–20 and 21–29 DAPs) and abiotic stress (cold, drought and salt). **b** Hierarchical clustering of rice HSF genes under heat stress in 14-day-old seedlings



showed much higher expression in all the tissues studied in this investigation (Table 2). However, differential expression pattern was also seen where intron number is same such as in the case of *OsHsfC2a* and *C2b*. Here no FL-cDNA was present for *C2b* showing negligible expression in all the tissues tested in contrast to the other gene in the pair, i.e. *C2a*. Further, the expression of different alternatively spliced variants was also studied by qRT-PCR. We found that in almost all the cases, the spliced variants had higher expression in all the tissues tested, particularly in case of *OsHsfA2a*, *A2d*, *B2a*, *B2b* and *B2c*. The higher expression levels of alternatively spliced variants could be due to the presence of extra regulatory elements in the additional intron. This could also be possible if a negative element gets deleted. Alternative splicing in *OsHSFA2a* results in the decrease of the length of the protein as well as the DBD in the fourth variant. Surprisingly, this gene model not only exhibits a major increase in expression level, but is detected in

additional stages as well. However, the third variant has an extra intron and shows variation in expression indicating the presence of some extra regulatory elements. A careful examination revealed that there are additional CCAAT and STRE elements present in the intron. This could be another reason of higher expression of the spliced variant under heat stress conditions. Similarly the higher expression levels of the second spliced form of *OsHSFA2d* can also be attributed to the presence of two extra HSEs located in the deleted intron. In case of *OsHSFB2b*, a difference of five introns in the two spliced forms not only results in a change in length of the total protein but also in an additional increase in the expression of the smaller protein. This emphasizes the presence of negative elements in the intron in this member of the *HSF* gene family. It is also imperative to mention here that the change in the length of DBD due to alternative splicing may be reflected in the downstream gene regulation. For example, for *OsHSFA2c*, the length of the protein as well as the DBD varies between the

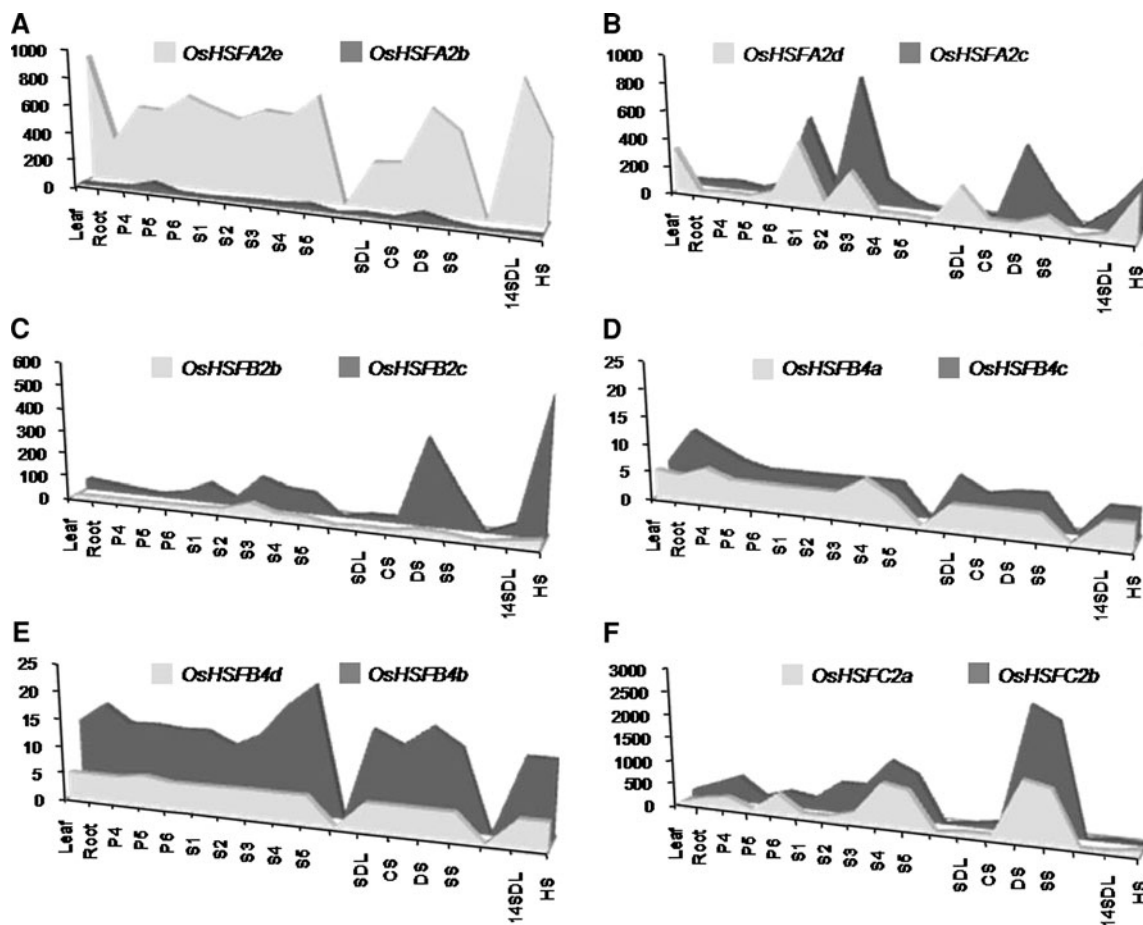


Fig. 4 Expression pattern of segmentally duplicated rice HSF genes in different growth stages as well as different abiotic stresses. (7-day-seedling (SDL), stages of panicle development (P4–P6) and, stages of

seed development (S1–S5) and cold stress (CS), drought stress (DS), salt stress (SS) and heat stress (HS) in 14-day-old seedling (14 SDL)

first and the fourth variants. However, their expression profiles remain the same. This difference may possibly affect the downstream genes of these two spliced forms which need to be examined further.

Similar to FL-cDNA/EST based expression profiling and microarray analysis, *OsHsfB4a* showed little or no change in expression in terms of relative fold change in qRT-PCR, suggesting that this gene (Os08g36700.1) is either not expressed or expresses at very low levels in some specialized tissues. However, this needs to be confirmed further by taking larger numbers of rice tissue samples.

Effect of short-term heat stress and calcium on HSF expression

The expression pattern of rice HSFs by exogenous calcium application with and without short-term heat stress was also studied. It was observed that 13 of the *OsHsf* genes are highly induced by calcium application (Table 3). Again these HSFs are also induced by short-term heat stress also, with a perfect correlation in calcium and short-term heat

stress. Interestingly, *OsHsfA2a* has exceptionally high expression change under heat stress, which is even higher with the combined effect of calcium and short-term heat stress. Second, we also find that whenever heat stress is combined with exogenous calcium treatment, the value of fold change increased dramatically except in the case of *OsHsfA2f* and *B2b*. *OsHsfA7* has little or no expression when calcium and heat stress are given separately but showed some expression when both these treatments are combined.

Analysis of regulatory elements in putative promoter regions

The presence of different abiotic stress-regulated *cis*-acting elements was analyzed by ELEMENT tool with inbuilt links to PlantCARE and PLACE database in the putative promoter region of rice HSF genes. Table 4 shows the description of some over represented motifs found by ELEMENT having links with previously described motifs in PLANTCARE and PLACE databases along with

Table 3 Effect of short time application of exogenous Ca^{++} and heat stress on expression analysis (relative fold change) of rice HSF genes

	CaCl_2	HS	HS + CaCl_2
<i>OsHsfA1a</i>	—	—	—
<i>OsHsfA2a</i>	13	13,711	21,559
<i>OsHsfA2b</i>	—	—	—
<i>OsHsfA2c</i>	6	531	632
<i>OsHsfA2d</i>	67	2,461	3,007
<i>OsHsfA2e</i>	17	20	39
<i>OsHsfA2f</i>	—	203	75
<i>OsHsfA3</i>	84	45	104
<i>OsHsfA4b</i>	—	—	7
<i>OsHsfA4d</i>	20	11	17
<i>OsHsfA5</i>	—	—	—
<i>OsHsfA7</i>	—	—	4
<i>OsHsfA9</i>	79	50	121
<i>OsHsfB1</i>	13	24	23
<i>OsHsfB2a</i>	10	347	403
<i>OsHsfB2b</i>	40	31	27
<i>OsHsfB2c</i>	121	155	186
<i>OsHsfB4a</i>	—	—	—
<i>OsHsfB4b</i>	—	—	—
<i>OsHsfB4c</i>	—	—	—
<i>OsHsfB4d</i>	—	—	—
<i>OsHsfC1a</i>	11	54	79
<i>OsHsfC1b</i>	—	—	—
<i>OsHsfC2a</i>	—	—	—
<i>OsHsfC2b</i>	—	—	—

Values are relative fold change in q-RT-PCR and—represents less than 1.5 fold change

corresponding Z score and *p* value. HSEs are found in all the heat-responsive *OsHsf* genes. We found that the characteristic heat shock element nGAAnnTTCnnGAAn (perfect HSE) is present in ten of the rice HSF genes (Table 4) and additionally imperfect HSE can be found in *OsHsf B1*, *B4c* and *C2b*. The presence of the HSE is perfectly correlated with the expression under heat stress in both micro-array and qRT-PCR analysis. We also found that almost all the rice HSF genes have one or more ABA-related motif in 1 kb putative promoter; this is also co-related with the expression of almost all the HSF genes by ABA application. Tissue-specific elements such as Skn-1 and GCN4, GATA conferring endosperm and pollen-specific expression were also found in many HSF genes and correlated well with the expression of *OsHsfA2a*, *A2c*, *A9*, *B2a* and *B2b* in the developing rice seed tissues. A low-temperature-responsive element, LTR, was also found in some HSF genes, especially in *OsHsfA3*, which has high expression during 2-h cold stress in both root and shoot tissues (Table 4). Among all the rice HSF genes, *A2a* has

maximum number of different *cis*-acting elements, especially related to heat and drought stress. Interestingly, this HSF is highly expressed in all the abiotic stresses and later stages of flower and seed development as revealed by microarray and real-time analysis. A close examination of its alternatively spliced forms revealed the presence of four HSEs in place of two, which can be correlated with the exceptionally high expression of this spliced variant under heat stress (Table 2). AGGGG motif termed as stress-responsive element (STRE) is present in almost all the rice HSFs except for *OsHsfA1* and *A4d*. Among all the rice HSF genes, *OsHsfA1* possess least number of tissue-specific and abiotic stress-responsive *cis*-acting elements in putative promoter region (Table 4), which is also reflected by the stable expression of this gene across all the tissues taken in the present study and in both micro-array and qRT-PCR expression analysis.

Discussion

HSFs are conserved proteins in all the living organisms; however, unlike animals, plants have a far greater number of HSF-coding genes than other organisms. While *Drosophila* and yeast have only one HSF and humans have four, most plants have multiple HSFs; for example *Arabidopsis* has 21 and rice has 25 members (Close et al. 1990; Sorger and Pelham 1988; Rabindran et al. 1991; Nover et al. 2001; Guo et al. 2008b). Search of EST database indicated the presence of at least 18 and 34 HSF gene members in tomato and soybean, respectively (Nishizawa et al. 2006). Our own analysis through BLASTN and BLASTP revealed the presence of 31 HSF gene models in *Populus trichocarpa* genome, which is further confirmed by Plant Transcription Factor Database (PlnTFDB 2.0).

The DNA-binding domain of plant HSF is encoded in two parts, separated by a single intron. The position of this intron is identical in all the HSFs; however, the size is variable (Nover et al. 2001). We found that rice is also not an exception and most of the *OsHsf* genes have only one intron in the DBD (Table 1). We also found that none of the rice HSF gene is intron less, which is at variance with the general observation that almost 20 percent of rice genes are intron less (Jain et al. 2008). Intron-less genes have been found in many rice transcription family genes such as MADS box (Arora et al. 2007), C2H2 zinc finger (Agarwal et al. 2007), bZIP (Nijhawan et al. 2008), SAUR (Jain et al. 2006) and F-box (Jain et al. 2007) gene families. Alternative splicing is known to change during environmental stresses, and at certain developmental stages, 10% of rice and 11.6% of *Arabidopsis* genes have been shown to be alternatively spliced across various tissues (Kikuchi et al. 2003; Lida et al. 2004). Among various rice HSFs, nine

Table 4 Analysis of putative *cis*-acting elements in 1-kbp upstream region of translation start codon

S no.	Motif	Z Score	<i>p</i> value	HSFs	Description
1	CAAAAAGA	6.06178	1.844e−06	A1a, A2a, A2b, A2f, A4d, A5, A7, B2c, B4c	Core site required for binding of DNA binding Dof proteins, with presumably only one zinc finger, and are unique to plants (Yanagisawa and Schmidt 1999)
2	ACTGTACG	7.55664	2.068e−14	A2e, B2c, B4d, C2a	Core of CuRE (copper-response element) involved in oxygen-response (Quinn and Merchant 1995)
3	CCGTTAC	3.60076	0.00015864	A3, B1, B2a, B4c	Binding site for two plant MYB proteins ATMYB1 and ATMYB2, both isolated from Arabidopsis, involved in regulation of genes that are responsive to water stress (Solano et al. 1995)
4	CCGACACG	6.25712	1.9608e−10	A2b, A2e, A3, A9, B2a	Core of low temperature responsive element (LTRE), ABA responsiveness, drought-induced gene expression (Baker et al. 1994)
5	nGAAnnTTCnnGAA	6.38424	8.6128e−11	A2a, A2b, A2c, A2d, A2e, A2f, A4b, B1, B2a, B2c	Heat shock element (HSE) found in the promoter of heat shock protein genes (Nover et al. 2001)
6	CGTGAGA	5.84144	5.402e−06	A2d, A4b, A4d, A9, B4a, B4c, C1a, C1b, C2a	GTGA motif found in the promoter of the pollen gene (Rogers et al. 2001)
7	AGGGG	4.52899	2.9633e−06	All except A1a and A4d	STRE (stress-responsive element) (Martinez-Pastor et al. 1996)
8	ACGTCCGC	5.26379	7.734e−05	A9, B1	ABA-responsive element (ABRE) of wheat Em (Em1a and Em1b) and rice rab21 genes (Marcotte et al. 1989)
9	ACGTGGC	6.03906	1.999e−06	All except A5, A7, B1, B2b, B4c	ABA-responsive element (ABRE) (Hobo et al. 1999)
10	GCGCTGAC	4.73338	1.104e−06	A3, A9, B4a,	Core of TGAC-containing W-box, binding site of rice WRKY71, a transcriptional repressor of the gibberellin signaling pathway, SA response (Eulgem et al. 1999)
11	CAGGTGGC	4.22742	1.1819e−05	A2c, A7, B4c, C2b	MYC recognition site found in the promoters of the dehydration-responsive gene rd22 (Abe et al. 2003)
12	CGAGAAAG	3.90095	4.7908e−05	A9, B2c, B4c	Responsible for pollen-specific activation (Filichkin et al. 2004)
13	CACACGTG	3.73255	9.4777e−05	A2c, C1b, C2b	Essential for expression of beta-phaseolin gene during embryogenesis in bean, tobacco, <i>Arabidopsis</i> (Chandrasekharan et al. 2003)
14	AGCAATTA	3.42727	0.00030484	A3, A4b, B2c	CCAAT box act cooperatively with HSEs to increase the hs promoter activity (Rieping and Schoffl 1992)
15	CCAGCTGG	3.26056	0.00055597	A2a, A3, A9	MYC recognition site found in the promoters of the dehydration-responsive rd22 (Abe et al. 2003)

genes showed predicted alternative spliced variants and they also show varied expression profiles as measured by qRT-PCR (Table 2). Intron-mediated enhancement (IME) of gene expression has been documented in several plant genes such as *Alcohol dehydrogenase 1* and *Bronze 1* (Callis et al. 1987), *Shrunken 1* (Clancy and Hannah 2002) in maize and *Phosphoribosylanthranilate transferase 1* in *Arabidopsis* (Rose and Last 1997). The enhancement of gene expression by introns has been related to transcription initiation, by increasing mRNA accumulation or by enhancing translational efficiency (LeHir et al. 2003).

In plants, HSFs are not only expressed during stress conditions but also in a tissue-specific manner (Almoguera

et al. 2002; Kotak et al. 2007; Swindell et al. 2007; Giorno et al. 2010). However, in the present study, we do not find much of tissue specificity of *OsHSF* gene; nevertheless, expression of several HSF genes was found appreciably higher in some tissues, especially panicle and developing seed (Table 2) such as *OsHsfA2a*, A3, A7 and A9. In *Arabidopsis*, it has been shown that HsfA9 is exclusively present in later stages of seed development and regulates the HSP genes during seed maturation (Kotak et al. 2007). There are constitutively expressed HSFs in plants, and these constitutive HSFs are proposed to have some role in house keeping functions and/or induction of other HSFs upon stress treatment such as *HSFA1a/A1b* in *Arabidopsis*

and *HsfA1a* in tomato (Busch et al. 2005; Mishra et al. 2002). In *Arabidopsis*, *HsfA1* is known as an early or constitutive expressed and primary regulator of heat stress (Wunderlich et al. 2007) and *HsfA2* is essential for prolonged heat stress, but not for the induction of thermotolerance (Charng et al. 2007). To test the synergistic effect of *AtHsfA1a* and *A1b*, Li et al. (2010) employed a GFP reporter system governed by heat shock gene promoters in the presence of effector plasmids containing *AtHsfA1a*, *A1b* and *A2* in protoplasts of *Arabidopsis*. A strong reporter gene activity was observed when both *AtHsfA1a* and *A1b* were present in combination, while the combination of *AtHsfA1a/A2* or *AtHsfA1b/A2* were less effective, again suggesting that *AtHsfA1* is primary inducer of HSR in *Arabidopsis*. Very recently, Liu et al. (2011) developed a quadruple mutant to remove functional redundancy of all *HsfA1* genes in *Arabidopsis*. It was found that more than 65% heat shock up-regulated genes were *HsfA1* dependent and apart from severely decreased thermo tolerance, the quadruple mutants have altered morphology and poor growth. In tomato also, it has been shown that localization of *HsfA2* to the nucleus requires interaction with *HsfA1* (Scharf et al. 1998) and over-expression of constitutive heat shock factor *HsfA1* confers thermo-tolerance via higher expression of *HsfA2* and *HsfB1*, resulting in the accumulation of several HSPs (Mishra et al. 2002). These evidences thus strongly suggest that *HsfA1* may act as a master regulator of heat stress response. On the basis of our expression data (Table 2), we also hypothesize that *OsHsfA1* may have similar function, as its expression is more or less stable in both microarray (Fig. 3) and real-time analysis across the tissues studied. Previously also it was shown that expression of rice *HsfA1* is least affected by various abiotic stresses such as heat, cold and salt and a stable expression was noticed at different time points under stress (Liu et al. 2010). It has been seen in *Arabidopsis* that expression of *HsfA2* was highest among all the class A HSFs under high temperatures and high light conditions (Nishizawa et al. 2006; Schramm et al. 2006). Rice has six members of *HsfA2*, and we found that expression of all the *OsHsfA2* genes is increased by heat stress except for *A2b*, which is actually a duplicated gene with *A2e* (Table 2; Fig. 4). In *Arabidopsis*, expression of *HsfA2* is increased within the first 15 min of stress. We also find a similar expression of *OsHsfA2a* in both early heat stress and exogenous calcium application (Table 3). Further, in *Arabidopsis* a total of six out of 21 HSFs were reported to be induced during heat stress (Busch et al. 2005). In rice we find almost the same set of HSFs induced by heat stress as 8 out of 25 *OsHsfs* genes are induced by heat stress (Table 2). Interestingly, all of these genes have HSEs in their putative promoter regions (Table 4) suggesting that these genes are actually activated by some other heat shock

factor which could be *OsHsfA1a*. However, this needs further experimentation for confirmation. Under non-stress conditions *AtHsfA1a* remain in inactive monomeric form, turning to trimeric active form by heat stress and binds to HSE in the promoters of HSF genes, thereby inducing thermo-tolerance (Lee et al. 1995; Zhang et al. 2001, 2003). Through chromatin immunoprecipitation (ChIP) Guo et al. (2008a, b) studied the putative DNA-binding motifs of *AtHsfA1a*. Apart from perfect and imperfect HSEs, STRE and TTC-rich elements were recognized as *AtHsfA1a*-binding sites with high affinity. The AGGGG motif termed as stress responsive (STRE) was identified in yeast and is responsive to various stresses (Martinez-Pastor et al. 1996). It was shown that deletion of STRE motif from the promoter of *Arabidopsis HSP90-1* gene decreases the promoter activity under heat stress conditions (Haralampidis et al. 2002). The AAAG motif has been suggested as target site for plant-specific Dof (DNA binding domain with one finger) class of transcriptions factors. Dof proteins have been found to interact with a variety of promoters related to seed storage, hormone and stress-responsive genes (Yanagisawa and Schmidt 1999). We found that this motif is present in all the abiotic stress-responsive rice HSFs. Interestingly, this is the only motif with a high Z score which is present in the putative promoter of *OsHsfA1a*.

As evident from the above discussion, class A HSFs have been characterized in more detail than class B and class C HSFs. Tomato *HsfB1* is a transcription co-activator functioning along with *HsfA1* and hypothesized as an HS-induced factor essential for maintenance and restoration of house keeping gene transcription during stress conditions (Bharti et al. 2004). In *Arabidopsis* *HsfB1* is heat inducible among all five class B HSF; however, its over-expression had no effect on HSP induction and thermotolerance (Prandl et al. 1998; Busch et al. 2005). In the present study, we observe that some class B HSFs are also induced by heat stress and express in developing seed tissues, especially *OsHsfB2a*, *B2b* and *B2c*. Recently, Kumar et al. (2009) developed a double knock-out mutant for *AtHsfB1* and *B2* and found that the double mutants have normal growth, fertility and thermo-tolerance as compared with single knock-out mutant or the wild type plants. However, the double mutant showed more resistance towards necrotrophic fungus and also showed higher expression of defensin genes *Pdf 1.2a* and *Pdf 1.2b*. Thus, it has been suggested that class B HSFs act as negative regulator of biotic stress response. ERF-associated amphiphilic repression (EAR) motif was identified in plants (Ohta et al. 2001), and it has been found that all five class B HSFs of *Arabidopsis* have the core region of EAR motif represented by R/KLFGV and the repressor activity of *AtHsfB1* and *B2b* was confirmed through Gal4BD system (Ikeda and

Ohme-Takagi 2009). We could also find the KLFGV motif in at least five of the class B HSF in rice viz. *OsHsfB1*, *B2a*, *B2b*, *B4a* and *B4b*.

Effect of calcium on heat stress signal transduction has been examined in earlier studies (Sangwan et al. 2002; Li et al. 2004; Liu et al. 2003, 2006; Chauhan et al. 2011). Available data suggest that heat stress elicits changes in the levels of intracellular Ca^{2+} , and Saidi et al. (2009) also reported the presence of a calcium channel in the plasma membrane, which opens only during the early phase of heat stress. Calmodulin (CaM) is a well-studied intra-cellular calcium sensor and it was shown that CaM is up-regulated during heat stress in maize and wheat (Gong et al. 1997; Liu et al. 2003). A CaM-binding protein kinase (*AtCBK3*) was identified which also interacts with *HSFA1a* during heat stress in *Arabidopsis* (Liu et al. 2008). Knock-out lines of this gene were found to have impaired basal thermo tolerance; however, over-expression of this gene results in higher thermo-tolerance through the increased binding activity of HSFs to HSP promoters. In the present study, we found that exogenous application of calcium increased the expression of several *OsHsfs*, *OsHsfA2a* showing remarkably high transcript accumulation (Table 3).

Conclusions

Plant Hsf family is one of the prime targets to study and elucidate the mechanism of abiotic stress response. Based upon expression profiling we hypothesize that *OsHsfA2a*, *A2d* and *B2a* are primary regulators of HSR in rice and *OsHsfA2a* plays important role during initial HSP gene induction during heat stress. This information can be used to produce stress-tolerant rice cultivars suitable for different stress environments. Transgenic tobacco plants have been developed by over-expressing a seed-specific, heat shock factor from sunflower *HaHsfA9* (Prieto-Dapena et al. 2008). The transgenic plants showed higher tolerance to drought stress in vegetative tissues and also have higher seed longevity. Heat- and salt-stress-tolerant transgenic *Arabidopsis* plants were also developed by over-expressing rice *HsfA2e* (Yokotani et al. 2008). Recently, *AtHsfA2* was over-expressed in *Arabidopsis* plants and the resultant transgenic plants showed higher anoxia tolerance (Banti et al. 2010). These studies thus provide evidence for the role of HSFs in abiotic stress-response pathway initiated not only in heat shock but also in other abiotic stresses.

Acknowledgments This work was financially supported by Department of Biotechnology, Government of India. HC and PA thank Council of Scientific and Industrial Research and the University Grants Commission, New Delhi, for the award of research fellowships during the course of this investigation.

References

- Abe H, Urao T, Ito T, Seki M, Shinozaki K, Yamaguchi-Shinozaki K (2003) *Arabidopsis* AtMYC2 (bHLH) and AtMYB2 (MYB) function as transcriptional activators in abscisic acid signaling. *Plant Cell* 15:63–78
- Agarwal P, Arora R, Ray S, Singh AK, Singh VP, Takatsuji H, Kapoor S, Tyagi AK (2007) Genome-wide identification of C_2H_2 zinc-finger gene family in rice and their phylogeny and expression analysis. *Plant Mol Biol* 65:467–485
- Almoguera C, Rojas A, Diaz-Martin J, Prieto-Dapena P, Carranco R, Jordano J (2002) A seed-specific heat-shock transcription factor involved in developmental regulation during embryogenesis in sunflower. *J Biol Chem* 277:43866–43872
- Arora R, Agarwal P, Ray S, Singh AK, Singh VP, Tyagi AK, Kapoor S (2007) MADS-box gene family in rice: genome-wide identification, organization and expression profiling during reproductive development and stress. *BMC Genomics* 8:242
- Baker SS, Withekn KS, Thomashow MF (1994) The 5' region of *Arabidopsis thaliana* cor15a has cis acting elements that confer cold-drought and ABA regulated gene expression. *Plant Mol Biol* 24:701–713
- Banti V, Mafessoni F, Loreti E, Alpi A, Perata P (2010) The heat-inducible transcription factor HsfA2 enhances anoxia tolerance in *Arabidopsis*. *Plant Physiol* 152:1471–1483
- Benjamini Y, Hochberg Y (1995) Controlling the false discovery rate: a practical and powerful approach to multiple testing. *J Roy Stat Soc Ser B* 57:289–300
- Bharti K, Schmidt E, Lyck R, Heerklotz D, Buback D, Scharf KD (2000) Isolation and characterization of HsfA3, a new heat stress transcription factor of *Lycopersicon peruvianum*. *Plant J* 22:355–365
- Bharti K, Von Koskull-Doring P, Bharti S, Kumar P, Tintschl-Korbitzer A, Treuter E, Nover L (2004) Tomato heat stress transcription factor HsfB1 represents a novel type of general transcription coactivator with a histone-like motif interacting with the plant CREB binding protein ortholog HAC1. *Plant Cell* 16:1521–1535
- Busch W, Wunderlich M, Schoff F (2005) Identification of novel heat shock factor-dependent genes and biochemical pathways in *Arabidopsis thaliana*. *Plant J* 41:1–14
- Callis J, Fromm M, Walbot V (1987) Introns increase gene expression in cultured maize cells. *Genes Dev* 1:1183–1200
- Chandrasekharan MB, Bishop KJ, Hall TC (2003) Module-specific regulation of the beta-phaseolin promoter during embryogenesis. *Plant J* 33:853–866
- Chang YY, Liu HC, Liu NY, Chi WT, Wang CN, Chang SH, Wang TT (2007) A heat-inducible transcription factor, HsfA2, is required for extension of acquired thermotolerance in *Arabidopsis*. *Plant Physiol* 143:251–262
- Chaudhary N, Nijhawan A, Khurana JP, Khurana P (2009) Carotenoid biosynthesis genes in rice: structural analysis, genome-wide expression profiling and phylogenetic analysis. *Mol Genetics Genomics* 283:13–33
- Chauhan H, Khurana N, Tyagi AK, Khurana JP, Khurana P (2011) Identification and characterization of high temperature stress responsive genes in bread wheat (*Triticum aestivum* L.) and their regulation at various stages of development. *Plant Mol Biol* 75:35–51
- Clancy M, Hannah LC (2002) Splicing of maize *Sh1* first intron is essential for enhancement of gene expression, and a T-rich motif increases expression without affecting splicing. *Plant Physiol* 130:918–929
- Close J, Westwood JT, Becker PB, Wilson S, Lambert U, Wu C (1990) Molecular cloning and expression of a hexameric

- Drosophila* heat shock factor subject to negative regulation. *Cell* 63:1085–1097
- Davletova S, Rizhsky L, Liang H, Shengqiang Z, Oliver DJ, Coutu J, Shulaev V, Schlauch K, Mittler R (2005) Cytosolic ascorbate peroxidase 1 is a central component of the reactive oxygen gene network of *Arabidopsis*. *Plant Cell* 17:268–281
- Eulgem T, Rushton PJ, Schmelzer E, Hahlbrock K, Somssich IE (1999) Early nuclear events in plant defence signalling: rapid gene activation by WRKY transcription factors. *EMBO J* 18:4689–4699
- Filichkin SA, Leonard JM, Monteros A, Liu PP, Nonogaki H (2004) A novel endo-mannanase gene in tomato *LeMAN5* is associated with anther and pollen development. *Plant Physiol* 134:1080–1087
- Giorno F, Wolters-Arts M, Grillo S, Scharf KD, Vriezen WH, Mariani C (2010) Developmental and heat stress-regulated expression of HsfA2 and small heat shock proteins in tomato anthers. *J Exp Bot* 61(2):453–462
- Gong M, Li YJ, Dai X, Tian M, Li ZG (1997) Involvement of calcium and calmodulin in the acquisition of heat-shock induced thermotolerance in maize. *J Plant Physiol* 150:615–621
- Guo J, Wu J, Ji Q, Wang C, Luo L, Yuan Y, Wang Y, Wang J (2008a) Genome-wide analysis of heat shock transcription factor families in rice and *Arabidopsis*. *J Genetics Genomics* 35:105–118
- Guo L, Chen S, Liu K, Liu Y, Ni L, Zhang K, Zhang L (2008b) Isolation of heat shock factor HsfA1a-binding sites in vivo revealed variations of heat shock elements in *Arabidopsis thaliana*. *Plant Cell Physiol* 49(9):1306–1315
- Haralampidis K, Milioni D, Rigas S, Hatzopoulos P (2002) Combinatorial interaction of *Cis* elements specifies the expression of the *Arabidopsis AtHsp90-1* gene. *Plant Physiol* 129:1138–1149
- Higo K, Ugawa Y, Iwamoto M, Higo H (1998) PLACE: A database of plant *cis*-acting regulatory DNA elements. *Nucleic Acid Res* 26:358–359
- Hobo T, Asada M, Kowyama Y, Hattori T (1999) ACGT-containing abscisic acid response element (ABRE) and coupling element 3 (CE3) are functionally equivalent. *Plant J* 19:679–689
- Ikeda M, Ohme-Takagi M (2009) A novel group of transcriptional repressors in *Arabidopsis*. *Plant Cell Physiol* 50(5):970–975
- Jain M, Tyagi AK, Khurana JP (2006) Genome-wide analysis, evolutionary expansion, and expression of early auxin-responsive SAUR gene family in rice (*Oryza sativa*). *Genomics* 88:360–371
- Jain M, Nijhawan A, Arora R, Agarwal P, Ray S, Sharma P, Kapoor S, Tyagi AK, Khurana JP (2007) F-box proteins in rice. Genome-wide analysis, classification, temporal and spatial gene expression during panicle and seed development, and regulation by light and abiotic stress. *Plant Physiol* 143:1467–1483
- Jain M, Khurana P, Tyagi AK, Khurana JP (2008) Genome-wide analysis of intronless genes in rice and *Arabidopsis*. *Funct Integr Genomics* 8:69–78
- Kikuchi S, Satoh K, Nagata T, Kawagashira N, Doi K, Kishimoto N, Yazaki J, Ishikawa M, Yamada H, Ooka H et al (2003) Rice Full-Length cDNA Consortium. Collection, mapping, and annotation of over 28,000 cDNA clones from japonica rice. *Science* 301:376–379
- Kotak S, Port M, Ganguli A, Bicker F, von Koskull-Döring P (2004) Characterization of C-terminal domains of *Arabidopsis* heat stress transcription factors (Hsfs) and identification of a new signature combination of plant class A Hsfs with AHA and NES motifs essential for activator function and intracellular localization. *Plant J* 39:98–112
- Kotak S, Vierling E, Baumelein H, von Koskull-Döring P (2007) A novel transcriptional cascade regulating expression of heat stress proteins during seed development of *Arabidopsis*. *Plant Cell* 19:182–195
- Kumar M, Busch W, Birke H, Kemmerling B, Nurnberger T, Schoff F (2009) Heat shock factor HsfB1 and HsfB2b are involved in the regulation of Pdf1.2 expression and pathogen resistance in *Arabidopsis*. *Mol Plant* 2:152–165
- Lee JH, Hubel A, Schoff F (1995) Derepression of the activity of genetically engineered heat shock factor causes constitutive synthesis of heat shock proteins and increased thermotolerance in transgenic *Arabidopsis*. *Plant J* 8:603–612
- LeHir H, Nott A, Moore MJ (2003) How introns influence and enhance gene expression? *Trends Biochem Sci* 28:215–220
- Li B, Liu HT, Sun DY, Zhou RG (2004) Ca²⁺ and calmodulin modulate DNA-binding activity of maize heat shock transcription factor in vitro. *Plant Cell Physiol* 45:627–634
- Li M, Berendzen KW, Schoff F (2010) Promoter specificity and interactions between early and late *Arabidopsis* heat shock factors. *Plant Mol Biol* 73:559–567
- Lida K, Seki M, Sakurai T, Satou M, Akiyama K, Toyoda T, Konagaya A, Shinozaki K (2004) Genome-wide analysis of alternative pre-mRNA splicing in *Arabidopsis thaliana* based on full-length cDNA sequences. *Nuc Acid Res* 32(17):5096–5103
- Liu HT, Li B, Shang ZL, Li XZ, Mu RL, Sun DY, Zhou RG (2003) Calmodulin is involved in heat shock signal transduction in wheat. *Plant Physiol* 132:1186–1195
- Liu HT, Gao F, Cui SJ, Han JL, Sun DY, Zhou RG (2006) Primary evidence for involvement of IP3 in heat-shock signal transduction in *Arabidopsis*. *Cell Res* 16:394–400
- Liu HT, Gao F, Li GL, Han JL, Liu DL, Sun DY, Zhou RG (2008) The calmodulin-binding protein kinase 3 is part of heat-shock signal transduction in *Arabidopsis thaliana*. *Plant J* 55:760–773
- Liu AL, Zou J, Zhang XW, Zhou XY, Wang WF, Xiong XY, Chen LY, Chen XB (2010) Expression profiles of class A rice heat shock transcription factor genes under abiotic stresses. *J Plant Biol* 53:142–149
- Liu H, Liao H, Chang Y (2011) The role of class A1 heat shock factors (HsFA1s) in response to heat and other stresses in *Arabidopsis*. *Plant Cell Environ*. doi:10.1111/j.1365-3040.2011.02278.x
- Martinez-Pastor MT, Marchler G, Schuller C, Marchler-Bauer A, Ruis H, Estruch F (1996) The *Saccharomyces cerevisiae* zinc finger proteins Msn2p and Msn4p are required for transcriptional induction through the stress response element (STRE). *EMBO J* 15:2227–2235
- Miller G, Mittler R (2006) Could heat shock transcription factors function as hydrogen peroxide sensors in plants? *Ann Bot* 98:279–288
- Mishra SK, Tripp J, Winkelhaus S, Tschiersch B, Theres K, Nover L, Scharf KD (2002) In the complex family of heat stress transcription factors, HsfA1 has a unique role as master regulator of thermotolerance in tomato. *Genes Dev* 16:1555–1567
- Mittler R, Merquiol E, Hallak-Herr E, Rachmilevitch S, Kaplan A, Cohen M (2001) Living under a “dormant” canopy: a molecular acclimation mechanism of the desert plant *Retama raetam*. *Plant J* 25:407–416
- Mockler TC, Michael TP, Preist HD, Shen R, Sullivan CM, Givan SA, McEntee C, Kay SA, Chory J (2007) The diurnal project: diurnal and circadian expression profiling, model based pattern matching and promoter analysis. *Cold Spring Harb Symp Quant Biol* 72:353–363
- Nijhawan A, Jain M, Tyagi AK, Khurana JP (2008) Genomic survey and gene expression analysis of the basic leucine zipper transcription factor family in rice. *Plant Physiol* 146:333–350
- Nishizawa A, Yabuta Y, Yoshida E, Maruta T, Yoshimura K, Shigeoka S (2006) *Arabidopsis* heat shock transcription factor A2 as a key regulator in response to several types of environmental stress. *Plant J* 48:535–547

- Nover L, Bharti K, Doring P, Mishra SK, Ganguli A, Scharf KD (2001) *Arabidopsis* and the heat stress transcription factor world: how many heat stress transcription factors do we need? *Cell Stress Chaperones* 6:177–189
- Ohta M, Matsui K, Hiratsu K, Shinshi H, Ohme-Takagi M (2001) Repression domains of class II ERF transcriptional repressors share an essential motif for active repression. *Plant Cell* 13:1959–1968
- Perez-Rodríguez P, Riano-Pachon DM, Correa LG, Rensing SA, Kersten B, Mueller-Roeber B (2009) PlnTFDB: updated content and new features of the plant transcription factor database. *Nucleic Acids Res* 38:D822–D827
- Prandl R, Hinderhofer K, Eggers-Schumacher G, Schoffl F (1998) HSF3, a new heat shock factor from *Arabidopsis thaliana*, derepresses the heat shock response and confers thermotolerance when overexpressed in transgenic plants. *Mol Genet* 258:269–278
- Prieto-Dapena P, Castano R, Almoguera C, Jordano J (2008) The ectopic overexpression of a seed-specific transcription factor, *HaHsfA9*, confers tolerance to severe dehydration in vegetative organs. *Plant J* 54:1004–1014
- Quinn JM, Merchant S (1995) Two copper-responsive elements associated with the *Chlamydomonas* Cyc6 gene function as targets for transcriptional activators. *Plant Cell* 7:623–628
- Rabindran SK, Giorgi G, Clos J, Wu C (1991) Molecular cloning and expression of a human heat shock factor, HSF1. *Proc Natl Acad Sci USA* 88:6906–6910
- Rieping M, Schoffl F (1992) Synergistic effect of upstream sequences, CCAAT box elements, and HSE sequences for enhanced expression of chimeric heat shock genes in transgenic tobacco. *Mol Gen Genet* 231:226–232
- Rizhsky L, Liang H, Mittler R (2002) The combined effect of drought stress and heat shock on gene expression in tobacco. *Plant Physiol* 130:1143–1151
- Rogers HJ, Bate N, Combe J, Sullivan J, Sweetman J, Swan C, Lonsdale DM, Twell D (2001) Functional analysis of cis-regulatory elements within the promoter of the tobacco late pollen gene g10. *Plant Mol Biol* 45:577–585
- Rombauts S, Dehais P, Van Montagu M, Rouze P (1999) PlantCARE, a plant cis-acting regulatory element database. *Nucleic Acid Res* 27:295–296
- Rose AB, Last RL (1997) Introns act post-transcriptionally to increase expression of the *Arabidopsis thaliana* tryptophan pathway gene PAT1. *Plant J* 11:455–464
- Saidi Y, Finka A, Muriset M, Bromberg Z, Weiss YG, Maathuis FJ, Goloubinoff P (2009) The heat shock response in moss plants is regulated by specific calcium-permeable channels in the plasma membrane. *Plant Cell* 21:2829–2843
- Sakuma Y, Maruyama K, Qin F, Osakabe Y, Shinozaki K, Yamaguchi-Shinozaki K (2006) Dual function of an *Arabidopsis* transcription factor DREB2A in water-stress-responsive and heat-stress-responsive gene expression. *Proc Natl Acad Sci USA* 103:18822–18827
- Sangwan V, Orvar BL, Beyerly J, Hirt H, Dhindsa RS (2002) Opposite changes in membrane fluidity mimic cold and heat stress activation of distinct plant MAP kinase pathways. *Plant J* 31:629–638
- Scharf KD, Heider H, Hohfeld I, Lyck R, Schmidt E, Nover L (1998) The tomato Hsf system: HsfA2 needs interaction with HsfA1 for efficient nuclear import and may be localized in cytoplasmic heat shock granules. *Mol Cell Biol* 18:2240–2251
- Schoffl F, Prandl R, Reindl A (1998) Regulation of the heat-shock response. *Plant Physiol* 117:1135–1141
- Schramm F, Ganguli A, Kiehlmann E, Englich G, Walch D, von Koskull-Doring P (2006) The heat stress transcription factor HsfA2 serves as a regulatory amplifier of a subset of genes in the heat stress response in *Arabidopsis*. *Plant Mol Biol* 60:759–772
- Schramm F, Larkindale J, Kiehlmann E, Ganguli A, Englich G, Vierling E, von Koskull-Doring P (2008) A cascade of transcription factor DREB2A and heat stress transcription factor HsfA3 regulates the heat stress response of *Arabidopsis*. *Plant J* 53:264–274
- Shim D, Hwang J-U, Lee J, Lee S, Choi Y, An G, Martinoia E, Lee Y (2009) Orthologs of class A4 heat shock transcription factor HsfA4a confer cadmium tolerance in wheat and rice. *Plant J* 21:4031–4043
- Singh G, Kumar S, Singh P (2003) A quick method to isolate RNA from wheat and other carbohydrate-rich seeds. *Plant Mol Biol Rep* 21:93a–93f
- Solano R, Nieto C, Avila J, Cañas L, Diaz I, Paz-Ares J (1995) Dual DNA binding specificity of a petal epidermis-specific MYB transcription factor (MYB.Ph3) from *Petunia hybrida*. *EMBO J* 14:1773–1784
- Sorger PK, Pelham HR (1988) Yeast heat shock factor is an essential DNA-binding protein that exhibits temperature-dependent phosphorylation. *Cell* 54:855–864
- Swindell WR, Huebner M, Weber AP (2007) Transcriptional profiling of *Arabidopsis* heat shock proteins and transcription factors reveals extensive overlap between heat and non-heat stress response pathways. *BMC Genomics* 8:125
- WRJr Marcotte, Russell SH, Quatrano RS (1989) Absciscic acid-responsive sequences from the *Em* gene of wheat. *Plant Cell* 1:969–976
- Wunderlich M, Doll J, Busch W, Kleindt CK, Lohmann C, Schoffl F (2007) Heat shock factors: regulators of early and late functions in plant stress response. *Plant Stress* 1:16–22
- Yamanouchi U, Yano M, Lin H, Ashikari M, Yamada K (2002) A rice spotted leaf gene, *Spl17*, encodes a heat stress transcription factor protein. *Proc Natl Acad Sci USA* 99:7530–7535
- Yanagisawa S, Schmidt RJ (1999) Diversity and similarity among recognition sequences of Dof transcription factors. *Plant J* 17:209–214
- Yokotani N, Ichikawa T, Kondou Y, Matsui M, Hirochika H, Iwabuchi M, Oda K (2008) Expression of rice heat stress transcription factor *OsHsfA2e* enhances tolerance to environmental stresses in transgenic *Arabidopsis*. *Planta* 227:957–967
- Yoshida T, Sakuma Y, Todaka D, Maruyama K, Qin F, Mizoi J, Kidokoro S, Fujita Y, Shinozaki K, Yamaguchi-Shinozaki K (2008) Functional analysis of an *Arabidopsis* heat-shock transcription factor HsfA3 in the transcriptional cascade downstream of the DREB2A stress-regulatory system. *Biochem Biophys Res Commun* 368:515–521
- Zhang L, Eggers-Schumacher G, Schoffl F, Prandl R (2001) Analysis of heat-shock transcription factor DNA binding in *Arabidopsis* suspension cultures by UV laser crosslinking. *Plant J* 28:217–223
- Zhang L, Lohmann C, Prandl R, Schoffl F (2003) Heat stress dependent DNA binding of *Arabidopsis* heat shock transcription factor HSF1 to heat shock gene promoters in *Arabidopsis* suspension culture cells in vivo. *Biol Chem* 384:959–963